Contents

List of Figures vii

List of Tables xiii

WBF Foreword xv

Foreword by Walt Boyes xvii

Preface xix

Introduction: ISA-88 Concepts Applied to Manufacturing Techniques Other Than Batch xxi

1 ISA-88 Isn't Just for Batches Anymore 1

2 Applying ISA-88 to Non-stop Production 11

3 Non-stop 88 Delivers Improved Operational Excellence 23

4 Taking ISA-88 Concepts to the Next Level 29

5 Experiences from Implementing a Single Control System Solution Using ISA-88 39

6 Using ISA-88 Batch Techniques to Manage and Control Continuous Processes 61

7 Using ISA-88 to Automate Procedures at Continuous Processing Facilities 73

8 Use of ISA-88 Techniques in Semi-continuous Applications 85

9 Integrating Quality and Process Information in a Batch Context for Semi-continuous Processes 95

10 Definition and Format of Recipes for the Packaging of Consumer Packaged Goods 101

11 Equipment Process Statecharts for Hybrid Manufacturing 119

12 Process Meets Discrete: Coupling of Batch Control to OMAC and to Real Packaging Lines 139

13 An Often Overlooked Aspect of Manufacturing and Packaging Systems Integration 149

14 Power Programming, ISA-88, and Packaging: Some of the Lessons Learned 171

15 Using ISA-88 to Define a Compliant Packaging Environment 187

16 Software Modularity: A Powerful Tool for Packaging Automation 201

17 Procedural Control and Exception Handling 211

Index 221

Figures

1.1.	ISA-88 and Make2Pack terminology.	3
1.2.	VFFS machine.	6
1.3.	VFFS EMs.	7
2.1.	Non-stop continuous production.	13
2.2.	NS88 rules.	15
2.3.	Sample non-stop production line.	16
2.4.	Sample recipe for a non-stop process cell.	16
2.5.	Time T3 product switchover in FILL #1.	18
2.6.	NS88 rules with reporting.	19
2.7.	Using reporting units to report on a batch.	20
2.8.	Buffering in a discrete process.	20
2.9.	Example of a typical equipment state model.	21
4.1.	End-to-end system representation of manufacturing.	32
4.2.	Example of standards-based controller logic.	33
4.3.	Generic and industry-specific representations of equipment hierarchy, based on ISA-95 part 3, draft 22.	34
4.4.	Conveyor configuration for gap generation.	35
4.5.	EM representation for coordination of conveyor system.	35
5.1.	Principal system structure diagram.	43
5.2.	Overall project execution.	44

5.3.	NNE project activity model with position of front-end definitions.	45
5.4.	Principle for building and process modules.	46
5.5.	Definition of computerized system.	46
5.6.	ISA-88 physical and procedural model used for FS development.	49
5.7.	Typical drawings with data for physical and procedural models (database objects).	50
5.8.	Structure of testing and qualification activities.	53
5.9.	Typical OQ test schedule applying to ISA-88 structure as key index.	56
5.10.	Contractual relationships.	59
6.1.	Simple batch example.	63
6.2.	Simple continuous example.	63
6.3.	Material delivery.	64
6.4.	Sequencer phase states.	67
6.5.	Material EM states and actions.	68
6.6.	The total system.	69
7.1.	Simplified distillation train.	77
7.2.	Distillation train simplified startup.	78
7.3.	Distillation train unit procedure identified.	80
7.4.	Distillation train operations identified.	81
7.5.	Distillation train sequence with ISA-88 applied.	82
8.1.	Simple three-column model.	88
8.2.	Three column model: Switching sequence.	89
8.3.	Building blocks of ISA-88.01.	90
8.4.	Cell and unit diagram.	91
8.5.	Column steps.	91
8.6.	On-line state steps.	92

8.7. On-line state with interrupts. 94

9.1. Integrating quality and process information. 99

10.1. Packaging pressures. 103

10.2. Business drivers versus packaging requirements. 106

10.3. Mode model. 108

10.4. State model for Automatic mode. 109

10.5. Process model for liquid filling. 111

10.6. Motion function blocks. 111

10.7. Physical model for a consumer packaged goods company. 112

10.8. Typical product hierarchy. 114

11.1. ISA-88.01 equipment state transition model for automatic control mode. 122

11.2. PackML state model version 3.0 for automatic control mode. 124

11.3. Typical material transfer pitfalls. 124

11.4. BasicControl enables definition of ProceduralControl. 126

11.5. BasicControl also enables definition of exception states. 127

11.6. ProceduralControl enables Producing. 127

11.7. Statechart of generic control. 128

11.8. ISA-88 equipment state transition model in statechart format. 129

11.9. Generic flow of Producing. 130

11.10. Extending the ISA-88 model toward an equipment process statechart. 133

11.11. Generic equipment manufacturing cycle. 135

11.12. Extending the Processing state for a specific equipment class cycle. 136

11.13. Example of equipment state manager pop-up. 137

12.1. Typical packaging line arrangement. 141

12.2. Example of the PLCopen camming functionality. 142

12.3. Example of a flow wrapper with synchronized multiple axes. 143
12.4. Positioning of the PLCopen motion control library over time. 144
12.5. Example of an implementation of the OMAC PackML state model. 145
12.6. Three solutions to winding. 146
12.7. An example of the mapping of the defined OMAC winding with constant velocity function block. 147
13.1. Machine-centric software approach. 151
13.2. Non-producing states due to equipment. 153
13.3. Non-producing states due to product flow. 153
13.4. Percentage of time in PackML states: Pre- and post-PackML state analysis. 154
13.5. Typical response times to equipment stoppages displayed in minutes. 155
13.6. Typical skills matrix expressed in percent proficiency. 155
13.7. Ideal fault finding solution expressed as skill set and required proficiency. 156
13.8. Initial MCN network diagram. 160
13.9. Threat probability and impact criticality rating. 161
13.10. Optimized MCN network diagram. 164
13.11. Fault finding process without remote connectivity. 166
13.12. Fault finding process with remote connectivity. 167
13.13. Current customer and manufacturing OEM relationship. 169
13.14. Potential customer and manufacturing OEM relationship. 170
14.1. From line shaft to servos. 172
14.2. Combining control and equipment to provide process functionality. 178
14.3. Separation of recipe and equipment control at the unit procedure. 179
14.4. VFFS machine. 179
14.5. Equipment decomposition. 180
14.6. Components of the Starting state. 181

14.7. Partial table of names for machine parameters. 182
14.8. Automatic state model diagram. 183
15.1. New blister ISA-88 procedural control model. 190
15.2. New blister physical model. 191
15.3. New blister ISA-88 process model. 192
17.1. State machine for handling procedural control exceptions. 216
17.2. Part of the PackML state machine, extracted from Figure 10.4. 216
17.3. Hierarchy of procedural control execution and state machines. 217

Tables

2.1.	Equipment state definitions	22
7.1.	Example batch sequences in continuous processing	75
8.1.	States and hold, restart, and abort handler actions	93
10.1.	A selection of tags	110

WBF Foreword

The purpose of this series of books from WBF, The Organization for Production Technology, is to publish papers that were given at WBF conferences so that a wider audience may benefit from them.

The chapters in this series are based on projects that have used worldwide standards—especially ISA-88 and 95—to reduce product variability, increase production throughput, reduce operator errors, and simplify automation projects. In this series, you will find the best practices for design, implementation, and operation and the pitfalls to avoid. The chapters cover large and small projects in a wide variety of industries.

The chapters are a collection of many of the best papers presented at the North American and European WBF conferences. They are selected from hundreds of papers that have been presented since 2003. They contain information that is relevant to manufacturing companies that are trying to improve their productivity and remain competitive in the now highly competitive world markets. Companies that have applied these lessons have learned the value of training their technical staff in relevant ISA standards, and this series provides a valuable addition to that training.

The World Batch Forum was created in 1993 as a way to start the public education process for the ISA-88 batch control standard. The first forum was held in Phoenix, Arizona, in March of 1994. The next few years saw growth and the ability to support the annual conference sessions with sponsors and fees.

The real benefit of these conference sessions was the opportunity to network and talk about or around problems shared by others. Papers presented at the conferences were reviewed for original technical content and lack of commercialism. Members could not leave without learning something new, possibly from a field thought to be unrelated to their work. This series is the opportunity for anyone unable to attend the conferences to participate in the information-sharing network and learn from the experiences of others.

ISA-88 was finally published in 1995 as *ISA-88.01-1995 Batch Control Part 1: Models and Terminology*. That same year, partially due to discussions at the WBF conference, ISA chartered ISA-95 to counter the idea that business people should be able to give commands to manufacturing equipment. The concern was that business

people had no training in the safe operation of the equipment, so boardroom control of a plant's fuel oil valve was really not a good idea. There were enough CEOs smitten with the idea of "lights-out" factories to make a firewall between business and manufacturing necessary. At the time, there was a gap between business computers and the computers that had infiltrated manufacturing control systems. There was no standard for communication, so ISA-95 set out to fill that need.

As ISA-95 began to firm up, interest in ISA-88 began to wane. Batch control vendors made large investments in designing control systems that incorporated the models, terminology, and practices set forth in ISA-88.01 and were ready to move on. ISA-95 had the attention of vendors and users at high levels (project-funding levels), so the World Batch Forum began de–emphasizing batch control and emphasizing manufacturing automation capabilities in general. This was the beginning of the transformation of WBF into "The Organization for Production Technology." Production technology includes batch control.

The WBF logo included the letters "WBF" on a map of the world, and since this well-known image was trademarked, the organization dropped the small words "World Batch Forum" entirely from the logo after the 2004 conference in Europe. WBF is no longer an acronym. Conferences continued annually until the economic crash of 2008. There was no conference in 2009 because many companies, including WBF, were conserving their resources.

WBF remained active and solvent despite the recession, so a successful conference was held in 2010 using facilities at the University of Texas in Austin. Several papers spoke of the need for procedural control for continuous and discrete processes. The formation of a new ISA standards committee (ISA106) to address this need was announced as well. Batch control is not normally associated with such processes, but ISA-88 has a large section on the design of procedural control. There is a need for a way to apply that knowledge to continuous and discrete processes, and some of those discussions will no doubt be held at WBF conferences, especially if the economy recovers. We would like to invite you to attend our conference and participate in those discussions.

WBF has always been an organization with an interest in production technologies beyond batch processing, even when it was officially "World Batch Forum." Over the years, as user interests changed, so has WBF. We have not lost our focus on batch; we have widened our view to include other related technologies such as procedural automation. We hope you will find these volumes useful and applicable to your needs, whatever type of process you have, and if you would like more information about WBF, we are only a simple click away at http://www.wbf.org.

William D. Wray, Chairman, WBF
Dennis L. Brandl, Program Chair, WBF
August 2010

Foreword by Walt Boyes

Many years ago, some dedicated visionaries realized that procedure-controlled automation would be able to codify and regularize the principles of batch processing. They set out on a journey that eventually arrived at the publication of the batch control standard ISA-88 and the development of the manufacturing language standard ISA-95.

Many end users have benefited from the work of these visionaries, who founded not only the ISA-88 Standard Committee but also the WBF. WBF has been an unsung hero in the conversion of manufacturing- to standards-based systems.

Today, WBF continues as the voice of procedure-controlled automation in the process and hybrid and batch processing industries. The chapters that make up this book series provide a clear indication of the power and knowledge of the members of WBF.

I have been proud to be associated with this group of visionaries for many years. *Control* magazine and ControlGlobal.com are and will continue to be supporters of WBF and its aims and activities.

I would like to invite you to come and participate in WBF, both online and at the WBF conferences in North America and Europe that are held annually. You will be glad you did. You can get more information at http://www.wbf.org.

Walt Boyes, ISA Fellow
Editor in Chief
Control magazine and ControlGlobal.com

Preface

This book is primarily about ISA-88.01 as it applies to non-batch control. The ISA Standards and Practices Board was asked to charter a batch control models and terminology group by Tom Fisher in 1988, and so the committee was called ISA88. I was active in that group from about 1990 to 1996. That was the golden age of standards, before mutual fund growth caused many in upper management to focus on growing the stock value via the bottom line and to lose track of what it took to maintain a manufacturing company. In other words, people couldn't get time for standards work or money for travel. It only got worse as time went on.

As a batch group with vendors, users, and consultants from that industry, we focused on batch processes. Only later did people begin to notice that we had created a pretty good set of engineering practices for *any* kind of automation of procedural control. We had tried to maintain a view of manual procedures during the deliberations of ISA88, but most people saw it as aimed at automation.

Many of the chapters in this book were written during the transition from ISA-88 as a batch standard to a procedural control standard. That name "batch" wouldn't let people see outside the box they'd put themselves in with a batch mindset. There is hope. We await the deliberations and results of ISA106, which has been chartered to address the needs of continuous and discrete processes.

It is ironic that process control, as it went through the pneumatic, electrical, and computerized stages, aimed first at the refineries. Once the refineries were happy, the vendor's growth opportunity was batch control. In the seventies, a DCS could be had with a Batch Package. Dr. Richard Lasher of Exxon called these packages "a FORTRAN compiler and a cheery 'good luck.'" ISA-88 brought a new order to batch control, causing users to demand better products and vendors to respond. Now that batch control systems make their customers happy (for the most part), vendors are looking at selling procedural control to the continuous and discrete markets.

ISA-88 will not help control system sales beyond Batch until vendors stop talking about Batch Engines, Managers, and Historians and replace Batch with Procedural Control (see Chapter 17).

Dave Chappell from P&G asked ISA to charter part 5 of ISA-88 for packaging machinery, which is a logical extension to batch control because many batch products require packaging to be sold. He also stirred up the Organization for Machine Automation and Control (OMAC), who created PackML and Make2Pack. Some people in these organizations, notably Adam Maki, found great value in ISA-88's exception handling state machine. They wanted to add some states (e.g., dividing Running into Producing and Standby) and needed to find a standards group that would let them do that. It really does the user no good to let each vendor come up with their own set of states and behavior. Think of the training issues.

Please read Dave Chappell's introduction to this book, which directly follows this Preface.

Chapters 1, 10, 12, and 14 address PackML and the needs of packaging machines. Chapter 10 by Ian McDonald of Unilever adds a business perspective to the discussion by one of the best in the business.

Chapters 2 and 3 discuss non-stop batch processing, as invented by Dennis Brandl. The process considers individually packaged foods as they move from one unit to the next, doing something different to the product in each stationary unit. Chapters 4, 6, 7, and 8 predict the effects of taking ISA-88.01 to the "next level," which means above batch to continuous and discrete processes. As a great fan of batch control, I consider continuous and discrete to be special cases of batch, and so they are a level down from batch control.

Chapter 5 is not concerned with the next level but with the construction of pharmaceutical plants and the extra needs of validation for federal regulations. It is well worth reading, whether your industry is regulated or not.

Chapter 9 discusses process information handling at the next level.

Chapter 11 covers statecharts for procedural control in exquisite detail. Willie Lotz is an isolated genius in my humble opinion, and so his work is not well known. His statecharts are definitive.

Chapter 13 adds network security and ISA-99 to the discussion. It is well worth reading if security is new to you.

Chapter 15 emphasizes compliance in a regulated industry for packaging.

Chapter 16 talks about the benefits of software modularity and introduces the concept of a Technology Evangelist.

I wrote chapter 17 to focus on the procedural control and exception handling that are needed for continuous and discrete control.

Bill Hawkins
Minneapolis, August 2010

Introduction

ISA-88 Concepts Applied to Manufacturing Techniques Other Than Batch

During the development of the ISA-88 batch standard in the late eighties, some of us on the committee evaluated the application of the batch concepts in this standard to determine whether or not they could effectively be applied to continuous and discrete manufacturing. The consensus was that the concepts fit very well and could be of valuable assistance to these other industries. Tom Fisher agreed but pointed out that if we tried to expand our charter beyond batch, the probability of us ever getting finished was greatly lessoned. Tom sure was right! The projected 2-year effort actually spanned 13 years to deliver ISA-88 parts 1 and 2 to satisfy the original charter. Good thing we stuck with batch, as I can't imagine where we'd be now if we had expanded our charter!

A few of us working in industries that used non-batch manufacturing did attempt to apply the ISA-88 concepts there. The results were quite interesting and taught many of us a lot about human nature and organizational power bases and momentum. To say the intrusion of this batch thinking was *not* well received in areas outside of batch would be a gross understatement. During the time of the ISA-88 development, batch was the most significant "pain" point for manufacturing. The other types of manufacturing were working just fine, thank you, so "don't mess with success" was the mantra of those organizations. Anyone who tried to evangelize the virtues of the ISA-88 batch concepts was quickly shown the door and was lucky if no bodily harm occurred! The current approaches for these manufacturing techniques were institutionalized, and the organizations that applied them were firmly in place and not open to new methods that might disrupt their comfortable lives. Understanding the ISA-88 concepts did require a little work to comprehend how to be successful in their application, and we learned that these other organizations were not willing to invest any effort in learning anything new.



As the ISA-88 concepts became institutionalized for batch processes, an interesting thing happened where there were both batch and continuous processes in manufacturing that had to work together. Having different approaches to delivering automation in a single environment created its own "pain," and the resolution that had the least pain associated with it was to select the ISA-88 concepts of modular equipment control and recipe procedural control and apply them to the total manufacturing process. The results were astounding, and those who were part of these efforts would attempt to carry the concepts back to their organizations, often with great success, for a while. This is where the organizational momentum and institutionalization of approaches that had been proven to be inferior surprised those of us in the batch segment of manufacturing. Over time spans that measured only a few years, the champions who were attempting to improve the continuous and discrete industries were moved out and the inferior institutionalized approach reasserted itself. To be fair, without the organizational support and acceptance of the concepts, adoption of anything new is doomed to failure. Business management seems to be clueless to the differences in approaches, so it is left up to the engineering disciplines to sort things out, which generally means "don't mess with success or take risks that might jeopardize a career."

This brings us to an interesting time in the early days of the new century. The packaging industry was struggling with issues that were requiring it to rethink how it delivered automation with packaging machinery. Working through the OMAC Packaging Workgroup (OPW), an effort to improve that industry was launched and championed by several large influential end user companies. This effort progressed with several deliverables; one of the most notable was the PackML guidelines Version 2 and Version 3, which provided guidelines for some very successful applications. Also during this time the ISA88 committee was evaluating the need to form a new working group to add more clarity on how to

deliver modular equipment automation. During a meeting in Orlando in February 2004, Rob Aleksa of Procter & Gamble approached the WBF and suggested a joint group be formed between the batch and packaging communities to evaluate if it was possible to jointly deliver something that would benefit both communities.

Thus the Make2Pack initiative was born, which I have led. (I started this as a Procter & Gamble employee and have continued to lead the effort after retirement.) This effort took over 2 years to complete with the investment of over 5000 effort hours involving 180 companies and 250 individuals. The results were a resounding affirmation that the two groups should work together and jointly create a global standard that would support batch and packaging. The ISA88 committee, under the leadership of Dennis Brandl, chartered the "Batch Control Part 5: Implementation Models and Terminology for Modular Equipment Control" working group, which got to work in 2006. During this effort, the group of packaging and batch practitioners identified and started work on several necessary additions to the effort that were outside of the scope of the charter. After reviewing the expansion to the part 5 charter, the ISA88 committee directed the part 5 group to refocus their efforts and remain within the original charter. They also authorized the part 5 group to develop a technical report that would address the needs of the packaging members of the committee.

The ISA technical report "TR88.00.05 Machine and Unit States" (an implementation example of ANSI/ISA-88.00.01) was started in 2006 and delivered to ISA in July 2008 for publishing. At that time ISA determined that the numbering was in error and changed the report to TR88.00.02. The report focused on how to use ISA-88 concepts to modularize a packaging machine and how to provide a new concept "Mode of Control" that provides multiple methods for managing a machine. The previous PackML guidelines only dealt with one Mode of Control, which was referred to as "Automatic." Using this term created a significant amount of misunderstanding in the early Make2Pack effort, as the term was used very differently from the way it is used by ISA-88. The technical report attempts to provide clarity and reduce confusion by adopting the term "Production" as the mode of control and not "Automatic." How helpful this will be remains to be determined.

Another milestone event during this time was that ISA-88 part 1 had been successfully applied for 10 years. As is customary with ISA/ANSI standards, 10 years after acceptance the standards are evaluated for continued relevance and, if relevant, for any updates that might be required. This evaluation led to an update effort that was led by Paul Nowicki, which was proposed to take only a few months. In the end, after several years, the part 1 update looks to be ready for acceptance. With the parallel efforts of part 5 and part 1 being so interactive and in the end dependant on one another, the decision was made by the new ISA-88 chairman, Randy Dwiggins, to suspend the part 5 work and focus on making sure

part 1 addressed any and all contentious issues part 5 had identified, which the part 1 update group has done. With the part 1 effort reaching completion, the part 5 effort is again progressing and will deliver proven automation approaches based on ISA-88 concepts that can benefit all manufacturing industries.

The adventure continues!

David A. Chappell
August 2010

ISA-88 Isn't Just for Batches Anymore

Presented at the WBF North American Conference, March 5–8, 2006, by

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Abstract

Applying the ISA-88 standard to continuous and discrete applications can save time and money. The ISA-88 modular concepts that lead to a structured design approach have gained a lot of momentum by providing cost savings in the batch process and packaging industries. Are batch processing and packaging the only industries that can benefit from standards and models? This chapter suggests that all control systems can benefit from the ISA-88 standard approach. System design and startup time reduction, easier long-term maintenance, and overall lower cost of ownership are benefits of a standard approach that apply to a wide variety of control applications. In conclusion, this chapter illustrates the potential time and cost savings attributable to applying the ISA-88 standards and models beyond batch processing applications.

Introduction

At the dawn of the ISA-88 standard, it was recognized that standard models and terminology were required before various batch control vendors and users could communicate. The lack of a standard resulted in many varied solutions to the batch control problem. These custom solutions were only understood by the individuals developing them. The ISA-88 standard has provided a vehicle for uniformly implementing batch solutions on different processes, at different sites, and even in different industries. By following the ISA-88 standard models, anyone familiar with the standard is able to look at any code and have a general idea of how it is structured and how to modify it if necessary.

The same issues exist in the continuous and discrete manufacturing industries. The Make2Pack effort recognizes this fact and is well on its way to finalizing conceptual models and methodologies that apply to the total manufacturing process. In addition, the ISA-88 committee will be issuing part 5 with the goal of applying ISA-88 to continuous and discrete processes as well. This chapter will illustrate how applying ISA-88 concepts can improve your discrete processes today.

How Can a Batch Standard Possibly Work in a Continuous or Discrete Application?

Whether one is looking at the physical model of batch, continuous, or discrete manufacturing, similar levels of equipment can be identified. At the lowest level there are simple devices such as valves, switches, and so on. These simple devices typically group to perform a higher level of functionality, and this higher functionality will ultimately produce a product. In the batch world, these are known as Control Modules (CMs), Equipment Modules (EMs), units, and process cells, respectively. Similar divisions can also be made in the continuous and discrete worlds.

The WBF Make2Pack effort has already begun to correlate terminology between batch, continuous, and discrete manufacturing. As seen in Figure 1.1, a slight change in the ISA-88 terminology makes it transferable to these other realms of manufacturing.

Why Use a Modular and Structured Programming Approach with Common Definitions?

Using common definitions and methodology across the batch, continuous, and discrete operations required to make a product eliminates confusion when translating

S88 part 1 term	Make2Pack term	Make2Pack Description
Process Cell	Production Line	A collection of one or more machines, linked together, to perform one or multiple tasks of the process for one or more products in a defined sequence. • Continuous Process (e.g. forming line in the food industry) • Converting Line (e.g. paper, fibers) • Discrete Manufacturing (e.g. assembly) • Packaging Line (from filling to secondary and tertiary packaging) S88: *A process cell contains all of the units, equipment modules, and control modules required to make one or more batches/lots.*
Unit	Unit Machine	In packaging, the unit corresponds to the logical grouping of mechanical and electrical assemblies that historically have been called machines. The term unit may apply to single function machine (filler, capper) or a multifunctional machine (monoblock filler/capper or any other configuration that combines functions within a single machine frame and control system). A multifunctional machine/unit can perform some or all functions of a packaging line, corresponding to process cell, that perform some or all of the functions of primary, secondary and tertiary packaging. A multifunctional machine may be logically broken down into several units corresponding to the individual functions. S88: *A unit is made up of equipment modules and control modules. The modules that make up the unit may be configured as part of the unit or may be acquired temporarily to carry out specific tasks.*
Control Module	Control Module	A Control Module is the lowest module in a physical model breakdown of a unit. The term Control Module relates to the combination of (a) physical equipment and the lowest level control component that controls this equipment to carry out a physical process action. There may be control modules without directly associated physical equipment. These control modules coordinate/supervise/sequence other control modules. NOTE: The use of the term control module to describe the supervisory/sequencing/coordinating functions is proving confusing and difficult to convey the concepts. Needs further consideration. Make2Pack examples of Control Modules: • Servo • Conveyor • Pneumatic Cylinder with feed-back S88: *A control module is typically a collection of sensors, actuators, other control modules, and associated processing equipment that, from the point of view of control, is operated as a single entity. A control module can also be made up of other control modules. For example, a header control module could be defined as a combination of several on/off automatic block valve control modules.*

Figure 1.1. ISA-88 and Make2Pack terminology.

meaning from one application to the next. A good example of the benefits of common definitions can be found in the following military example. If one branch of the Norwegian Military were to pass the command "secure the building" on to another branch, they would each interpret that command differently. The army would surround it, provide cover fire by snipers and heavy machine guns, and then clear the building room by room. The air force would make sure the windows were closed, the lights turned off, and the doors properly locked at the end of the day. The navy would sign a 10-year lease contract, with options to buy it after 5 years. Each of these actions are appropriate and meet the needs of the individual branches of the military but could bring disastrous results if the command was given from one branch to another.

The same is true for the steps in a manufacturing process. To improve manufacturing output as product is passed from one production step to the next, information must flow faster and transition times must be reduced. As companies continue to connect the top floor to the shop floor, "secure the product" will need to mean the same thing in every step of the process. When the business management system sends a command to the manufacturing floor, all processes and machines must have the same understanding of that command to ensure that the command is executed properly and safely.

Demands on Manufacturing

What demands are placed on today's manufacturing facilities? The same that have always been—today's manufacturing facilities are asked to provide an ever-increasing return on investment for the products they manufacture. To achieve this end, the team is challenged to continually make product faster, better, and cheaper. How does following the ISA-88 standard models help reach these goals? Let's examine each individually:

1. Faster

- *Reduced design and startup time*. By reusing modules of code, design time and the time required to start up and troubleshoot a new machine are reduced.
- *Greater process uptime*. When consistent code is used throughout the manufacturing processes, operations and maintenance workers can more easily find and understand the root cause of a maintenance issue.

2. Better

- *Improved quality*. Using standard terminology to define the components and processes involved in manufacturing, Quality Assurance is able to define consistent metrics for continuously improving the quality of the product.
- *Improved validation*. Common code reduces the time required to validate changes, as a change can be validated once and used in many different places.

3. Cheaper

- *Less initial capital investment*. Common code modules allow manufacturers to reuse code, enabling shorter design cycles and startup time and ultimately less up-front investment.
- *Less training*. Since the same concepts and code are used, operators and maintenance personnel do not need to be retrained every time a new process or machine is installed in their plant.
- *Improved efficiency*. Consistently defined states throughout all modules of equipment allow for quick identification of which modules are not being fully utilized.

Applying ISA-88 principles can help achieve the continuous improvement goals that all manufacturing facilities must reach. The next section illustrates such improvements when applying ISA-88 standards to non-batch operations.

A Real-World Example of a Discrete System Using ISA-88 Standards

The following is an example of implementing a discrete machine using the ISA-88 standards and concepts. The application is from the packaging industry and is known as a Vertical Form, Fill, and Seal (VFFS) machine, as illustrated in Figure 1.2. The machine consists of four main operations: weigh, bag form, feed roll, and seal and cut. In other words, the machine weighs out some product, forms a bag or package around the product, feeds it through, and cuts and seals the package.

The Original Equipment Manufacturer (OEM) broke the machine down into EMs and CMs as suggested by the ISA-88 standard. As illustrated in Figure 1.3, each of the four main operations of the machine was configured as an EM. The elements and devices within each EM were defined as CMs. The entire machine

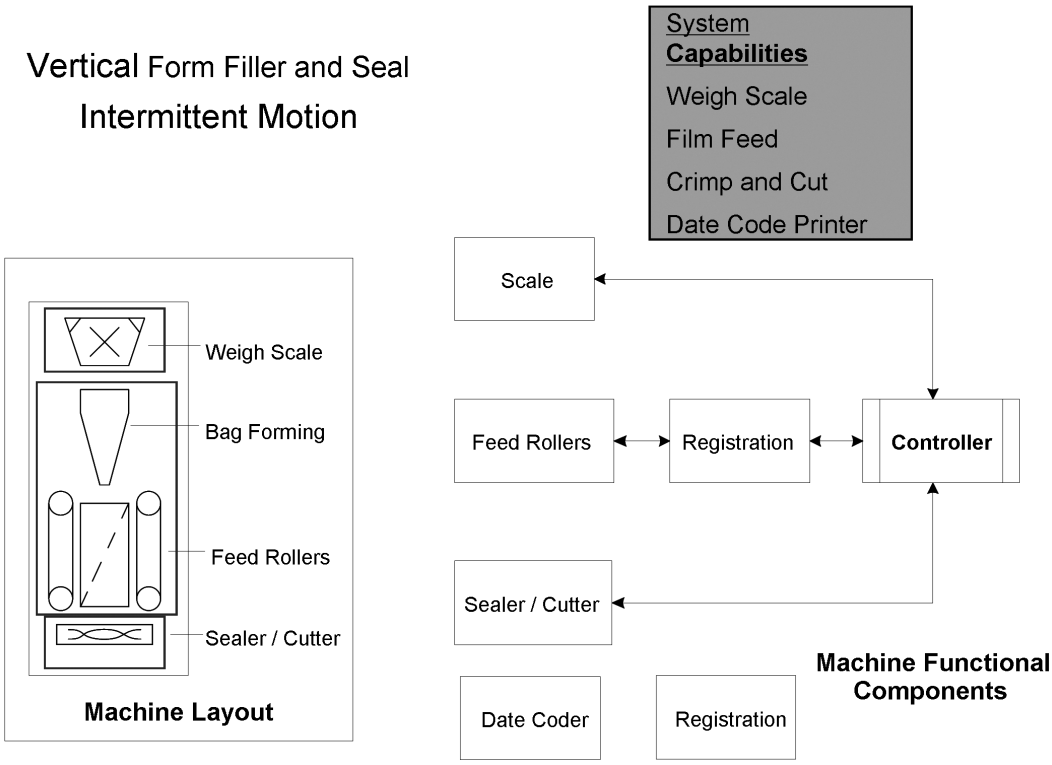


Figure 1.2. VFFS machine.

was defined as a unit. By defining the machine in this way, the OEM was able to create modular code that could be reused within this machine and also on other applications.

The OEM was then able to structure the code into modules that could utilize a modified ISA-88 state model as the foundation. This made the code easier to write and understand because there was a model to follow. The unit (machine) program transitioned according to rules set forth by the state model. The state model could then be used to develop a functional specification, programming document, training document, user document, and Human Machine Interface (HMI).

Program Design Methodology

The program design methodology consisted of the following:

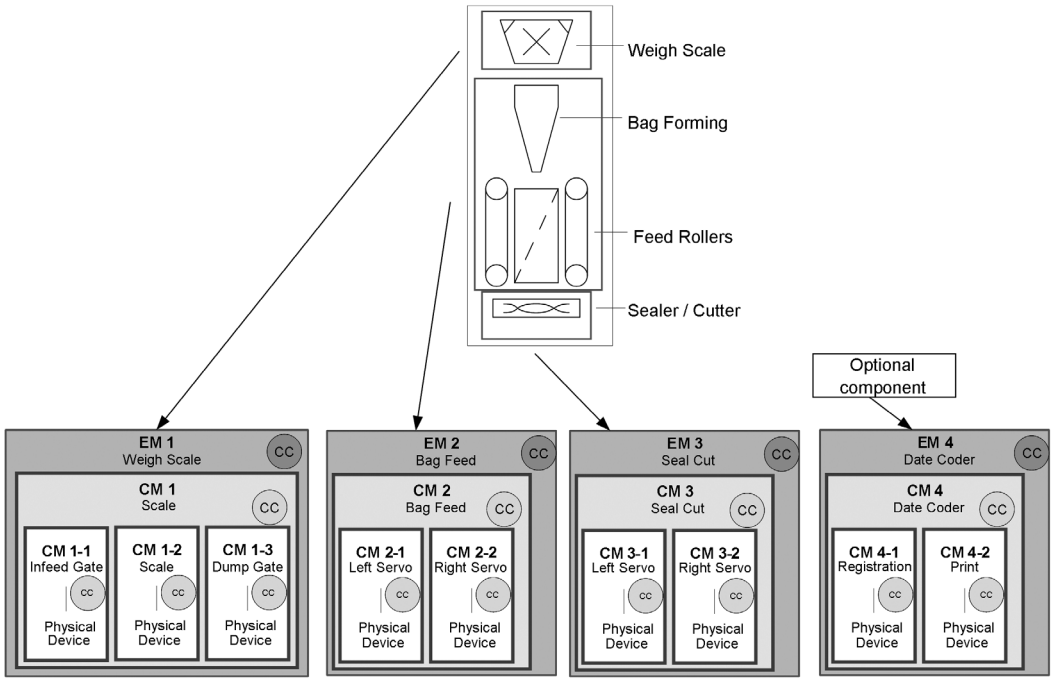


Figure 1.3. VFFS EMs.

- Define Physical Model
  - Break the unit (machine) into Functional Components and Sub-components (EMs and CMs)
- Define Equipment Procedural Model
  - Drive the complexity as low as possible into the CMs
  - Make interface to CMs as simple as possible
- Populate program

Program the CMs and EMs

With the Physical and Procedural Models defined, the OEM programmed nine CMs for this machine. Five of the modules were servo devices, so the OEM was able to duplicate the code for these, which simplified the program structure. By following this structure, the OEM and the end user only needed to understand one servo device module as opposed to five modules (each coded slightly differently

because they did not share common states and functionality as defined by ISA-88). The OEM used a specific algorithm to start, stop, reset, gear, and cam the servos. With one CM being defined for servo devices, the OEM was able to consistently apply this algorithm to all of the servos.

In addition, the scale vendor provided some basic control code that was designed to ISA-88 principles (the scale vendor also supplies the batch industry). This code easily "plugged into" the code and design philosophy of the machine as an EM with three CMs. The scale vendor defined the functions of the scale system with the complexity of its function buried in the details of the "black box" control code. The interface was easily understood and clearly defined as inputs, outputs, status, and so forth. The OEM felt the scale was easy to add to his machine. In reality, a scale may be extremely complex, with multiple scales and an averaging algorithm. All this was transparent to the OEM.

The OEM designed the date coder section of the code from scratch. This code included control for one servo for registration correction and a printer. The printer was defined as the final CM, with logic for the basic control of the printer and for sending ASCII strings that contained the codes to be printed on the product.

Define Linkages between Modules

The OEM next considered the linkages between the base CMs and their respective EMs. In the servo devices' CM there is interlocking for electronic camming and other servo-only level functions that are in the CM code but not exposed to the end user. The cam timing and parameter values can be accessed, but access to the algorithms for starting and stopping is of no value to the end user. The end user of the machine may only care about the system starting and stopping smoothly and the cycle time. The OEM can do its magic by delivering a machine that performs optimally within the user guidelines. By creating cam profiles that maximize the acceleration and deceleration within the required product cycle time, value may be added to the machine by delivering smoother operation and lower shock on the machine and product.

The OEM considered how the CMs would be accessed and what data would be transferred in and out of the module. In this way the interface to the CM was as clean and intuitive as possible.

This effort made "plugging" the CM code into the EM simple because the interface points were clearly defined. By coding this module and testing it in one application, the OEM was confident that the code would work when plugged into other parts of the machine or into other machines.

Programming the Operation of the Machine

At this point the code was set up to represent the physical structure of the machine. Next, the OEM looked at the operations that the VFFS machine needed to perform. In this case the machine needed to do the following:

- Thread up the machine with packaging materials
- Start up production of the machine
- Empty out the machine
- Manually jog one section

When designing the control code for a machine using ISA-88 concepts and methodologies, at the highest level of machine control one considers the physical and procedural models and how they interact together in a machine to provide useful functions. From an ISA-88 perspective these functions correspond to Unit Procedures or Operations for the machine. The OEM set up the functions of a machine as phases. From this level, the OEM considered which phases of the machine to invoke. An individual phase such as threadup or many phases may run simultaneously (or as a recipe) while producing a product.

Benefits Derived from Using the ISA-88 Standard

By following the ISA-88 standard, the OEM would be able to reuse this code for next-generation machines or other packaging applications. For example, the OEM was able to use the weigh scale EM without modification for a different machine that also included a weigh scale. Likewise, the OEM would be able to reuse the servo device CM and printer CM when appropriate. Codes that are structured to fit the ISA-88 models are powerful because they can be separated at different levels (i.e., unit, EM, and CM) and plugged into other applications where appropriate because of their common definitions and structures. This avoids large blocks of monolithic code that are difficult to understand and even harder to debug and maintain.

The OEM has also found that their field support personnel are now able to quickly understand the machine code and have expressed appreciation, as the modules have found their ways onto other machines. The field support personnel no longer have to understand how control engineer Frank programs versus control engineer Ted. Now Frank and Ted use the same structure by following the ISA-88 principles. If Frank and Ted were authors, they would both write their books with

the same structural components: title, table of contents, chapters, paragraphs, and sentences, and possibly even an index and a glossary.

From a maintenance perspective there is a benefit for both the OEM and the end user. By following the standard, the customer is able to easily understand the structure of the code, eliminating the time required to troubleshoot issues when they arise. In addition, if the customer performs batching operations they may have other ISA-88 based installations within their facility. Since the code is easy to understand, the customer is less likely to make a mistake when modifying the code, which minimizes the support costs for the OEM. If there is an issue it should be easy to locate, as the module of code associated with the device in question is defined as a specific CM.

In an ideal world the end user would never need to modify code once a machine is delivered; however in the real world one must be prepared for such an event. Using models and terminology as defined by ISA-88 was key in helping this OEM successfully manage changes.

As this OEM continues to roll the ISA-88 standard into other applications, it believes it will reduce design time, startup time, installation time, support trips, and end user training time. These savings will significantly reduce its overall cost of designing a machine while improving the quality of the end product. This standard allows the OEM and its end users to meet the goal of making the product faster, better, and cheaper.

Conclusion

The previously stated examples illustrate real improvements and cost savings to a discrete manufacturing process by implementing ISA-88 principles. In addition to the immediate efficiency gains on the manufacturing floor, a consistent and commonly defined programming structure enables a connection to business-level systems by enforcing the use of common definitions and methodologies.

One can see that the ISA-88 standard isn't just for batches anymore. ISA-88 concepts can have the same value for continuous and discrete processes as well. Future efforts from the Make2Pack and ISA88 committees will continue to evolve and improve the ability to implement structured programming concepts and methodologies. These efforts will provide a standard and consistent programming structure across automation at the process, site, and enterprise levels.